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A comprehensive survey of machine and deep learning as well as explainable artificial intelligence approaches for heart disease detection

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Abstract

Cardiovascular diseases (CVDs) remain the foremost cause of mortality worldwide, claiming approximately 17.9 million lives annually and representing 32% of all global deaths. Timely and accurate detection of heart disease is therefore of paramount importance. This paper presents a narrative survey, synthesising 57 peer-reviewed studies (2000–2025, concentrated in 2020–2025) of computational methods for heart disease detection, encompassing classical machine learning, deep learning, hybrid architectures, explainable artificial intelligence, and emerging paradigms such as federated learning and IoT-integrated systems. We reviewed studies covering diverse data modalities (clinical tabular data, ECG signals, echocardiograms, cardiac MRI, chest X-rays, and wearable sensor streams), benchmark datasets, feature engineering and selection strategies, performance evaluation metrics, and clinical deployment considerations. The survey finds that ensemble methods such as XGBoost and Random Forest report accuracies ranging from the low 80 s to as high as 98% on structured clinical datasets, and that convolutional neural network (CNN), long short-term memory (LSTM), and transformer-based models report comparably wide ranges (roughly 86–99.7%, by AUC and accuracy respectively) on ECG and imaging benchmarks. Critically, we find that the studies reporting the highest figures are almost exclusively evaluated on small, single-centre benchmark datasets without external or prospective validation, so these numbers should be read as a ceiling on benchmark performance rather than as evidence of clinical-grade accuracy; several of the reviewed studies show an explicit training-to-test accuracy gap consistent with overfitting. We further discuss explainability techniques such as SHAP, LIME, and Grad-CAM that are considered important for clinical trust, and identify open challenges including class imbalance, data scarcity, limited external validation, reproducibility, multi-modal fusion, model generalisability, and fairness.

Keywords Heart disease detection, Cardiovascular disease, Machine learning, Deep learning, ECG classification, Explainable AI, Clinical decision support, Survey

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1 Introduction

Cardiovascular diseases (CVDs) constitute the leading cause of death globally, responsible for an estimated 17.9 million fatalities per year—approximately 32% of all deaths worldwide [1]. In the United States alone, heart disease incurred approximately $252.2 billion in direct and indirect costs between 2019 and 2020 [2]. Projections by the World Health Organization indicate that CVD-related deaths will rise from 20.5 million in 2025 to 35.6 million by 2050, representing a 90% increase over 25 years [3].

The umbrella term *cardiovascular disease* encompasses a spectrum of disorders including coronary artery disease (CAD), myocardial infarction (MI), heart failure, arrhythmias, valvular disease, and congenital heart defects. These conditions share the critical characteristic that their progression is often subclinical for prolonged periods, with overt symptoms appearing only at advanced stages. Early detection substantially improves survival rates, reduces treatment costs, and enables lifestyle modifications that can halt or reverse disease progression [4].

Historically, cardiologists relied on clinical examinations, resting electrocardiograms (ECGs), echocardiography, blood biomarkers (troponin, BNP), and coronary angiography for diagnosis. These approaches—while clinically effective—are time-consuming, expensive, operator-dependent, and often unavailable in resource-limited settings. The explosion of electronic health records (EHRs), wearable monitoring devices, and medical imaging, combined with advances in machine learning (ML) and deep learning (DL), has created an unprecedented opportunity to automate, accelerate, and democratise heart disease detection [5].

This survey provides a structured review of the computational landscape for heart disease detection, addressing:

i. Data sources, publicly available datasets, and preprocessing pipelines;
ii. Classical ML methods and their benchmark performance;
iii. Deep learning architectures including CNNs, RNNs, LSTMs, and transformers;
iv. Hybrid and ensemble strategies;
v. Explainability and interpretability methods;
vi. IoT, wearable, and federated learning systems;
vii. Challenges, limitations, and future research directions.

1.1 Scope and methodology

This is a narrative (non-systematic) survey: it does not follow a registered PRISMA protocol with a quantified screening flow, and this limitation is made explicit here so that readers can calibrate their expectations of the evidence base accordingly.

Search strategy. Candidate literature was identified through keyword searches of IEEE Xplore, PubMed/PMC, Scopus, ScienceDirect, SpringerLink, Nature/Scientific Reports, and arXiv/medRxiv preprint servers, using combinations of the terms *heart disease*, *cardiovascular disease*, *machine learning*, *deep learning*, *ECG classification*, *explainable AI*, *federated learning*, and *wearable/IoT cardiac monitoring*. Reference lists of retrieved reviews were additionally hand-searched (backward snowballing) to identify primary studies and dataset descriptor papers.

Inclusion/exclusion criteria. We included peer-reviewed journal and conference articles, and a small number of widely cited preprints, published mainly between 2020

and 2025 (with earlier foundational and dataset papers retained where they remain the standard reference, e.g. the UCI Cleveland and MIT-BIH datasets). Studies were included if they reported a computational method (classical ML, DL, hybrid, or XAI) applied to heart-disease-related detection, classification, or risk prediction, with a quantified evaluation metric (accuracy, AUC, F1, or equivalent). Studies were excluded if they addressed cardiovascular conditions only tangentially, lacked a quantitative evaluation, or were not accessible in full text.

Corpus. Applying this process yielded the 57 references synthesised in this survey, comprising primary technical studies, dataset/benchmark descriptor papers, epidemiological sources, and a small number of prior reviews [52–56] used for cross-checking coverage and identifying gaps (Sect. 10 tabulates the reported metrics extracted from the primary technical studies). Because inclusion was not exhaustive and no independent dual-screening was performed, the corpus should be read as an illustrative, expert-curated sample of the field rather than a complete census of the literature; this is consistent with the survey's aim, which is synthesis and critical appraisal rather than meta-analytic estimation.

1.2 Paper organisation

The remainder of this survey is organised as follows. Section 2 provides a clinical overview of heart disease types. Section 3 surveys datasets and preprocessing techniques. Section 4 covers feature engineering and selection. Section 5 reviews classical ML methods. Section 6 details deep learning approaches. Section 7 discusses hybrid and ensemble strategies. Section 8 addresses explainability. Section 9 examines IoT and federated systems. Section 10 presents a consolidated performance comparison. Section 11 offers a critical synthesis addressing generalisability, evaluation validity, architecture trade-offs, and translational readiness. Section 12 identifies open challenges. Section 13 outlines future directions. Section 14 concludes the survey.

2 Clinical background: types of heart disease

2.1 Coronary artery disease

Coronary artery disease (CAD) arises from atherosclerotic plaque accumulation in the coronary arteries, reducing blood supply to the myocardium. CAD is the most prevalent form of heart disease and the primary driver of myocardial infarction. Clinical indicators include chest pain (angina), shortness of breath, and ST-segment changes on ECG. ML models have targeted CAD prediction using risk factors such as age, sex, cholesterol levels, blood pressure, diabetes, and smoking history [6].

2.2 Myocardial infarction

Acute myocardial infarction (AMI), commonly referred to as a heart attack, results from sudden occlusion of a coronary artery, causing ischaemic necrosis of the myocardium. Early reperfusion therapy (percutaneous coronary intervention or thrombolysis) is highly time-sensitive. Automated ECG analysis with deep learning has shown promise in real-time STEMI (ST-elevation MI) detection [7].

2.3 Heart failure

Heart failure (HF) denotes the inability of the heart to pump adequate blood to meet the body's metabolic demands. It is characterised by reduced ejection fraction (HFrEF) or preserved ejection fraction (HFpEF). Biomarkers such as NT-proBNP, imaging parameters such as left ventricular ejection fraction (LVEF), and longitudinal EHR data are commonly used in ML-based HF prediction and risk stratification [8].

2.4 Cardiac arrhythmias

Arrhythmias encompass a wide spectrum of abnormal cardiac rhythms, including atrial fibrillation (AF), ventricular tachycardia (VT), ventricular fibrillation (VF), and atrioventricular blocks. AF alone affects more than 33 million people globally and substantially elevates stroke risk. Automated ECG classification with CNNs and transformers has reported accuracy approaching cardiologist-level agreement for AF detection on benchmark datasets [9], though, as discussed in Sect. 11.1, this has not yet been confirmed through independent external validation.

2.5 Valvular heart disease

Valvular disorders—including aortic stenosis, mitral regurgitation, and tricuspid regurgitation—result in abnormal blood flow patterns detectable through echocardiography and phonocardiography (PCG, or heart sounds). Deep learning models processing PCG signals have reported accuracy exceeding 99% in identifying valvular pathologies on the benchmark datasets studied [10], a figure that, per Sect. 11.2, should be interpreted with appropriate caution given the small size of these datasets.

2.6 Congenital heart disease

Congenital heart defects (CHDs) are structural abnormalities present at birth, ranging from simple septal defects to complex univentricular hearts. The epidemiology of CHDs has evolved significantly, with improved surgical outcomes increasing the adult CHD population [11]. ML tools are increasingly used for post-operative risk stratification.

3 Data sources and datasets

3.1 Structured clinical datasets

3.1.1 UCI cleveland heart disease dataset

The Cleveland dataset, originally collected at the Cleveland Clinic Foundation and hosted on the UCI Machine Learning Repository, is the most widely used benchmark in heart disease prediction research [12]. It contains 303 patient records with 76 raw attributes, though researchers conventionally use the 13 most clinically relevant features plus a binary target (presence/absence of heart disease). Despite its small size, it remains the de facto standard for comparing ML classifiers. Known limitations include missing values in the *thal* and *ca* features, and limited demographic diversity [13].

3.1.2 UCI statlog (heart) dataset

Derived from the Cleveland dataset, this 270-sample dataset uses identical features and is commonly paired with Cleveland in comparative studies [14].

3.1.3 Hungarian, Switzerland, and long beach VA datasets

These three additional UCI subsets, when combined with Cleveland, yield a multi-centre dataset of 1,190 cases used by several studies to improve generalisability [15].

3.1.4 Framingham heart study dataset

The longitudinal Framingham Heart Study dataset captures risk factors over time, enabling survival analysis and time-to-event modelling. Researchers have used it to predict 10-year coronary heart disease risk using logistic regression, decision trees, and gradient boosting [16].

3.1.5 Kaggle heart disease datasets

Several Kaggle-hosted datasets—often aggregations of the above UCI subsets extended with additional clinical variables—have been adopted for research due to their accessibility and larger sample sizes (up to 1,190+ records with 11–15 features) [13].

3.1.6 MIMIC-III and MIMIC-IV

The Medical Information Mart for Intensive Care (MIMIC) databases contain rich, longitudinal EHR data from tens of thousands of ICU patients, enabling complex risk modelling for in-hospital cardiac events, decompensation, and mortality [8].

3.2 ECG signal datasets

3.2.1 MIT-BIH arrhythmia database

The MIT-BIH dataset contains 48 two-lead Holter ECG recordings, each 30 min long, with expert annotations for 19 rhythm classes. It is the most widely used benchmark for arrhythmia classification algorithms [17].

3.2.2 PTB-XL

The PTB-XL dataset provides 21,837 12-lead ECG recordings (10 s each) from 18,885 patients with 71 ECG statement classes. Its scale and multi-label nature make it ideal for evaluating deep learning classifiers [18].

3.2.3 PhysioNet challenge datasets

The annual PhysioNet Computing in Cardiology Challenges have released multiple publicly available ECG datasets for tasks including AF detection, cardiac risk stratification, and 12-lead ECG classification [19].

3.2.4 CPSC 2018

The China Physiological Signal Challenge 2018 dataset contains 12-lead ECG recordings annotated for nine rhythm/morphology categories, fostering cross-population model evaluation [20].

3.3 Imaging datasets

3.3.1 EchoNet-dynamic

A dataset of 10,030 apical-4-chamber echocardiography videos labelled with LVEF values, enabling regression and classification of systolic function using video-based CNNs [21].

3.3.2 Cardiac MRI datasets

Datasets such as the ACDC (Automated Cardiac Diagnosis Challenge) and UK Biobank cardiac MRI repository support segmentation and functional assessment of cardiac chambers with deep learning models [22].

3.3.3 ChestX-ray14

A dataset of 112,120 frontal chest radiographs from 30,805 patients labelled for 14 pathologies including cardiomegaly, used for multi-label classification with CNN transfer learning [23].

3.4 Preprocessing techniques

Effective preprocessing is essential for high-quality model training. Common steps include:

- **Missing value imputation:** Mean/median imputation, k-nearest-neighbour imputation, and multiple imputation by chained equations (MICE) are widely applied to structured clinical datasets.
- **Normalisation and scaling:** Min-max normalisation and z-score standardisation are applied to continuous clinical features such as serum cholesterol and resting blood pressure.
- **Class imbalance handling:** The Synthetic Minority Over-sampling Technique (SMOTE), ADASYN, SMOTETomek, SMOTEENN, and under-sampling clustering are employed to address skewed class distributions prevalent in real-world cardiac datasets [24].
- **ECG denoising:** Bandpass filtering, wavelet denoising, and empirical mode decomposition (EMD) are applied to remove baseline wander, power-line interference, and muscle artefacts from ECG signals.
- **Image augmentation:** For imaging datasets, random flips, rotations, brightness adjustments, and Gaussian noise injection prevent overfitting and improve generalisation.

4 Feature engineering and selection

4.1 Clinical feature categories

Research on structured clinical data has identified several dominant feature categories for heart disease prediction:

- **Demographic:** Age, sex, body mass index (BMI).
- **Haemodynamic:** Resting blood pressure, maximum heart rate achieved.
- **Biochemical:** Serum cholesterol, fasting blood glucose.
- **Electrocardiographic:** Resting ECG results, ST depression (oldpeak), ST slope, exercise-induced angina.
- **Imaging-derived:** LVEF, wall motion score, left atrial volume.
- **Symptomatic:** Chest pain type, number of major vessels coloured by fluoroscopy, thalassemia type.

4.2 Feature selection methods

Dimensionality reduction and feature selection are critical when working with high-dimensional clinical or genomic data:

- **Filter methods:** Chi-square test, mutual information, SelectKBest, and F-statistic are computationally efficient and model-agnostic [25].
- **Wrapper methods:** Recursive Feature Elimination (RFE) iteratively removes the least important features, guided by a base estimator.
- **Embedded methods:** LASSO regularisation and tree-based feature importance (from Random Forest or XGBoost) simultaneously train the model and select features.
- **Evolutionary methods:** Genetic algorithms (GA), Particle Swarm Optimisation (PSO), and Social Group Optimisation (SGO) have been applied to optimise feature subsets for cardiac risk prediction [26].

4.3 ECG feature extraction

For ECG signals, feature engineering transforms raw waveforms into informative representations:

- **Time-domain features:** RR interval, QRS duration, QT interval, P-wave amplitude, T-wave morphology.
- **Frequency-domain features:** Power spectral density, heart rate variability (HRV) metrics in low-frequency (LF) and high-frequency (HF) bands.
- **Time-frequency features:** Short-time Fourier transform (STFT), wavelet transform coefficients, and empirical mode decomposition (EMD) [50].
- **Non-linear features:** Sample entropy, approximate entropy, Lyapunov exponents, Poincaré plot metrics.
- **Learned features:** End-to-end CNN models learn hierarchical features automatically, bypassing manual extraction.

5 Classical machine learning methods

A broad body of work applies standard supervised learning pipelines, comparing algorithm families on the same benchmark features to identify a practical baseline for heart disease prediction [57].

5.1 Logistic regression

Logistic regression (LR) is a foundational probabilistic classifier modelling the log-odds of heart disease as a linear function of input features. Despite its simplicity, LR achieves competitive accuracy of 80–86% on the UCI Cleveland dataset and remains a clinically interpretable baseline. LR is also the foundation of the Framingham Risk Score, a widely used clinical tool for 10-year CVD risk estimation [16].

5.2 Support vector machines

Support vector machines (SVMs) construct an optimal separating hyperplane in a kernel-transformed feature space. The radial basis function (RBF) kernel is most commonly applied. Studies on the Cleveland dataset report SVM accuracy of 80–92%, with the range depending heavily on preprocessing and hyperparameter tuning [6]. SVM

has been praised for its robustness in high-dimensional spaces but is computationally expensive on large datasets.

5.3 Decision trees and random forest

Decision trees provide inherently interpretable classifications through hierarchical feature splits. However, they are prone to overfitting on small datasets like Cleveland. Random Forest (RF), an ensemble of decision trees trained on bootstrapped subsets with random feature subsets, substantially mitigates overfitting. RF consistently ranks among the top performers for clinical heart disease prediction, achieving 88–94% accuracy on standard benchmarks [27]. Studies using ROC-AUC analysis show RF achieving 0.92-–0.95, reflecting excellent discriminative power [15].

5.4 Gradient boosting methods

5.4.1 XGBoost

XGBoost (Extreme Gradient Boosting) builds an ensemble of shallow trees sequentially, each correcting residual errors from the previous. It has emerged as the most frequently cited best-performing classical classifier in multi-model comparative studies. On the combined UCI dataset, XGBoost achieves 93% accuracy with ROC-AUC of 0.94 [15], while cross-validation confirms sustained performance of 91–93% on Cleveland [14].

5.4.2 LightGBM and CatBoost

LightGBM uses gradient-based one-side sampling and exclusive feature bundling for faster training on large datasets. CatBoost handles categorical features natively without encoding. Both achieve accuracy within 1–2% of XGBoost with significantly reduced computational requirements [6].

5.4.3 AdaBoost

AdaBoost adaptively weights misclassified instances, yielding strong performance on imbalanced cardiac datasets. Combined with LIME-based explainability, AdaBoost achieved 99.33% accuracy on the Heart Attack Classification Dataset [24].

5.5 Naïve bayes

Gaussian Naïve Bayes models feature distributions as conditionally independent Gaussians given the class. Despite this strong independence assumption, it achieves 78–83% accuracy on Cleveland and has been used in lightweight, real-time mobile cardiac screening applications [6].

5.6 K-nearest neighbours

K-nearest neighbours (KNN) classifies instances by majority vote among the k nearest training samples in feature space. KNN achieves 87–91% accuracy on Cleveland, with hyperparameter tuning yielding optimal k values of 5–11 [28].

5.7 Artificial neural networks

Shallow multi-layer perceptrons (MLPs) with 2–4 hidden layers and sigmoid or ReLU activations have been applied to clinical data, reaching accuracies of 85–96% in some

studies [6]. However, MLPs are outperformed by modern deep architectures when sufficient data is available.

5.8 Comparative performance on UCI cleveland

Table 1 summarises representative accuracy values from recent comparative studies on the UCI Cleveland benchmark.

6 Deep learning approaches

6.1 Convolutional neural networks

Convolutional neural networks (CNNs) have become the dominant architecture for processing structured spatial or temporal signals in cardiac diagnosis. Their capacity to learn hierarchical, translation-invariant feature representations directly from raw data eliminates the need for manual feature engineering.

6.1.1 1D CNNs for ECG classification

One-dimensional CNNs operate directly on raw ECG waveforms. Dutta et al. [29] proposed an efficient 1D-CNN for CAD prediction achieving 95.7% accuracy on the UCI dataset. Mehmood et al. [30] demonstrated 97.6% accuracy on PTB ECG data using deep 1D-CNNs with batch normalisation. The MIT-BIH benchmark has seen CNN-based classifiers achieve over 99% accuracy in five-class arrhythmia classification [31].

6.1.2 2D CNNs for ECG images and cardiac imaging

ECG signals can be converted to scalogram images using continuous wavelet transforms (CWT) or plotted as raw signal images, enabling 2D-CNN processing. For echocardiography, the EchoNet-Dynamic CNN regresses LVEF with mean absolute error of 4.1%, comparable to inter-observer cardiologist variability [21]. CNN-based DenseNet and EfficientNet architectures for phonocardiogram classification have achieved 99.35-–99.65% accuracy in valvular disease detection [10].

6.1.3 Transfer learning

Pre-trained CNNs (ResNet, VGG, InceptionNet, EfficientNet, DenseNet) trained on large natural image datasets (ImageNet) are fine-tuned for cardiac imaging tasks. Maran [47] developed CNN-based models analysing multimodal data (BMI, ECG, PTB), achieving 98% accuracy in heart disease prediction. Transfer learning has been especially impactful when labelled cardiac imaging data is scarce.

Table 1 Comparative ML performance on the UCI Cleveland Heart Disease Dataset

Method	Accuracy (%)	Precision (%)	Recall (%)	ROC-AUC
Logistic regression	80–86	81–84	78–88	0.86–0.90
SVM (RBF Kernel)	80–92	80–90	79–91	0.88–0.93
Naïve bayes	78–83	77–82	76–85	0.83–0.87
K-nearest neighbour	87–91	86–90	88–92	0.90–0.93
Decision tree	78–85	78–84	79–87	0.80–0.87
Random forest	88–94	89–93	90–94	0.92–0.95
Gradient boosting (GBM)	87–92	86–91	87–92	0.90–0.93
XGBoost	91–93	91–92	91–92	0.93–0.95
LightGBM	88–92	88–91	88–92	0.91–0.94
Artificial neural net	84–96	84–95	84–96	0.89–0.97

6.2 Recurrent neural networks and LSTM

Recurrent neural networks (RNNs) and their gated variants—long short-term memory (LSTM) and gated recurrent unit (GRU)—are designed for sequential data modelling. Their ability to retain long-range temporal dependencies makes them well-suited for ECG time-series analysis.

Alamatsaz et al. [33] proposed a lightweight hybrid CNN-LSTM for detecting eight arrhythmia classes, achieving 97.8% accuracy on MIT-BIH. Rai et al. [34] applied a hybrid CNN-LSTM to ECG big data for automatic arrhythmia prediction. GAN-LSTM ensemble models for ECG classification have addressed class imbalance and achieved 0.994 AUC on the PTB-ECG dataset [35].

6.3 Transformer-based models

The success of attention mechanisms and transformer architectures in NLP (BERT, GPT) has inspired their adaptation to ECG and cardiac data. Transformers excel at capturing long-range temporal dependencies without the sequential bottleneck of RNNs.

6.3.1 HeartBEiT

HeartBEiT is a vision-based transformer pre-trained on 8.5 million ECGs using masked image modelling (BEiT framework). Fine-tuned for STEMI detection, hypertrophic cardiomyopathy, and low LVEF classification, it outperforms ResNet-152 and EfficientNet-B4 at all training data fractions, demonstrating 28.4% improvement at 1% training size [32].

6.3.2 HCTG-Net

The Hybrid CNN-Transformer Network with Gated Fusion (HCTG-Net) combines a residual CNN for local waveform patterns with a transformer branch for global temporal context. A learnable gated fusion mechanism adaptively balances contributions. On MIT-BIH, HCTG-Net achieved 0.9946 accuracy and 0.9711 F1-score, outperforming prior CNN-only and LSTM-only methods [31].

6.3.3 CTRhythm

CTRhythm integrates CNN local feature extraction with transformer global contextual learning for atrial fibrillation detection from single-lead ECG, addressing the limitations of CRNN-based methods on long sequences [9].

6.4 Autoencoder and generative models

Autoencoders compress cardiac data into latent representations for anomaly detection and dimensionality reduction. Saleh Alghamdi et al. [38] combined autoencoder and DenseNet architectures on the UCI Cleveland dataset, achieving a mean accuracy of 99.67% and test accuracy of 99.99%. Generative adversarial networks (GANs) have been applied to synthesise realistic ECG data for data augmentation, addressing the chronic scarcity of labelled pathological recordings. The Wolf-based GAN system (WbGAS) classified ECG into normal sinus rhythm, arrhythmia, and congestive heart failure with high accuracy [37].

6.5 Attention mechanisms

Self-attention mechanisms, applied as stand-alone modules within CNN or LSTM architectures, enable models to weight the contribution of individual ECG beats or image regions differently. Channel attention (Squeeze-and-Excitation networks) and spatial attention have improved echocardiography segmentation accuracy on ACDC datasets.

6.6 Multi-modal deep learning

Combining heterogeneous data modalities—ECG signals, imaging, clinical tabular features, and genomics—through late fusion or cross-modal attention has yielded superior performance to any single modality. Maran [47] demonstrated that multimodal CNNs (BMI + ECG + PTB) outperform unimodal baselines by 5–8%.

7 Hybrid and ensemble strategies

7.1 Ensemble methods

Ensemble learning combines predictions from multiple base models to reduce variance (bagging), bias (boosting), or both.

Voting ensembles combine classifier outputs by majority vote or weighted averaging. Almulihi et al. [39] demonstrated that deep learning-based voting ensembles outperform individual models for early heart disease prediction (AUC = 0.95).

Stacking trains a meta-learner on base-model outputs. A conditional variational autoencoder with stacked ensemble learning achieved superior performance on myocardial infarction detection compared to single-model baselines [40].

7.2 Hyperparameter optimisation

Bayesian optimisation, genetic algorithms, and metaheuristic methods have been used to tune model hyperparameters:

- The Social Group Optimisation (SGO) algorithm applied to RF and XGBoost improved accuracy from 84% to 93%+ on Cleveland [14].
- Adaptive Elephant Herd Optimisation (AEHOM) trained a multi-layer deep CNN (MLDCNN) achieving top benchmark accuracy [26].
- Tunicate Swarm–Sail Fish Optimisation for channel selection in wireless body area network (WBAN) cardiac monitoring.

7.3 Hybrid CNN-classical ML pipelines

Several studies combine deep feature extraction from CNNs with classical ML classifiers:

- CNN features fed to SVM or RF classifiers often outperform either component alone on ECG and imaging datasets.
- Bharti et al. [41] compared hybrid ML/DL models on supervised learning coronary heart disease datasets, finding deep learning achieved 94.2% average accuracy.

7.4 Multi-stage prediction systems

Three-stage WBAN architectures pipeline data aggregation, channel selection, and ML classification, achieving robust real-time cardiac risk assessment in energy-constrained wearable contexts.

8 Explainable artificial intelligence (XAI)

8.1 Motivation for explainability

Despite impressive accuracy metrics, the adoption of complex ML and DL models in clinical cardiology has been hampered by their "black-box" nature. Clinicians require justifiable, transparent predictions to build trust and integrate AI recommendations into clinical workflows. Regulatory frameworks including the EU AI Act and FDA guidance on Software as a Medical Device (SaMD) increasingly mandate explainability [24].

8.2 SHAP (SHapley additive explanations)

SHAP assigns each feature a Shapley value representing its marginal contribution to a specific prediction, grounded in cooperative game theory. Key advantages include:

- Both global (dataset-level) and local (instance-level) explanations.
- Consistent across diverse model classes (trees, neural networks, linear models).
- Identifies that *age*, *maximum heart rate*, *chest pain type*, and *ST depression* are consistently among the most influential features for heart disease prediction across multiple studies [42, 43].

Rezk et al. [42] applied SHAP to XGBoost/LightBoost ensembles for heart disease prediction, making risk factors visible and clinically interpretable.

8.3 LIME (Local interpretable model-agnostic explanations)

LIME approximates complex model predictions locally with an interpretable surrogate model. It generates perturbed instances around the input of interest and fits a simple linear model, identifying features most responsible for the individual prediction [42]. For IoMT-integrated heart disease systems, LIME-derived explanations have been shown to align with clinical expert judgement, enabling practical clinical decision support [44].

8.4 Grad-CAM and attention visualisation

For imaging and ECG-image models, Gradient-weighted Class Activation Mapping (Grad-CAM) highlights image regions most influential to a CNN prediction. HeartBEiT demonstrated that transformer attention maps localise biologically relevant ECG waveform segments for STEMI and HCM diagnosis, offering superior explainability over standard CNNs [32].

8.5 XAI-HD framework

The XAI-HD hybrid framework integrates ML, DL, and XAI techniques, systematically addressing class imbalance and missing data through advanced preprocessing (OSS, NCR, SMOTEN, ADASYN, SMOTETomek, SMOTEENN). Evaluated across three clinical heart disease datasets (CHD, FHD, SHD), XAI-HD reduces classification error rates by 20–25% versus traditional ML-only models [24].

8.6 Clinical adoption of XAI

Studies show that cardiologists are more likely to accept AI diagnostic support when explanations align with their clinical reasoning. XAI methods have been evaluated in real clinical settings, showing improved physician confidence and reduced over-reliance on model outputs.

9 IoT, wearables, and federated learning

9.1 IoT-integrated cardiac monitoring

The proliferation of Internet of Things (IoT) devices has enabled continuous, real-time cardiac monitoring outside hospital settings. Key system components include:

- **Wearable ECG patches and smartwatches:** Devices such as the Apple Watch, AliveCor KardiaMobile, and ZIO patch continuously record single-lead or multi-lead ECGs for ambulatory arrhythmia detection.
- **Photoplethysmography (PPG) sensors:** Wrist-worn PPG sensors enable AF detection and HRV analysis without traditional ECG leads.
- **Electronic stethoscopes:** CNN-integrated electronic stethoscopes (e.g., Raspberry Pi 4B + EfficientNet-B3) achieve 99.35% accuracy in valvular disease screening at point of care [46], and broader cyber-physical systems have been proposed to support valve disease screening and diagnosis end-to-end [51].
- **Cloud-based telehealth:** IoT-enabled ECG monitors transmit signals to cloud inference engines, enabling physician notification in cardiac emergencies [35].

9.2 Wireless body area networks (WBANs)

WBAN-based heart disease prediction models aggregate multi-sensor physiological data, optimise channel selection with metaheuristic algorithms (Tunicate Swarm, Sail Fish Optimisation), extract weighted entropy features, and classify risk using ensemble ML, achieving robust real-time performance with low communication overhead.

9.3 Federated learning

Federated learning (FL) enables collaborative model training across multiple healthcare institutions without sharing raw patient data, addressing critical privacy regulations (HIPAA, GDPR). In FL for cardiac risk prediction:

- Each hospital trains a local model on its own patient data.
- Only model parameters (gradients or weights) are shared with a central aggregator.
- The global model benefits from diverse multi-centre populations.

Recent studies demonstrate that FL models approach centralised training performance within 2–5% accuracy, while providing strong privacy guarantees. Key challenges include heterogeneous data distributions across sites (non-IID data), communication efficiency, and adversarial robustness [45].

9.4 Edge computing for cardiac inference

Deploying ML/DL models on edge devices (smartphones, Raspberry Pi, FPGAs) enables low-latency cardiac inference without cloud dependency. Model compression techniques including pruning, quantisation, and knowledge distillation are essential for fitting DL models within edge hardware constraints.

10 Performance comparison

Table 2 summarises representative deep learning and hybrid method performance across multiple modalities.

Table 2 Representative deep learning and hybrid model performance across cardiac data modalities. Where a study reports several settings (e.g. two datasets, or pre-/post-optimisation), the range or the specific setting is given rather than a single rounded figure, and validation protocol and sample size are reported alongside accuracy so that results can be compared on a like-for-like basis

Study/Model	Modality	Dataset (*n*)	Val.	Task	Acc. (%)	AUC
HCTG-Net [31]	ECG signal	MIT-BIH (∼100k beats)	Held-out	Arrhythmia, multi-class	99.46	0.998
HeartBEiT [32]	ECG image	8.5M ECGs (pre-train)	External	MI / low-EF	–	0.86–0.95
CNN-LSTM [35]	ECG signal	PTB-ECG	Held-out	Disorder class.	99.4	0.995
CNN-BiLSTM [36]	ECG signal	MIT-BIH	Held-out	Arrhythmia class.	98.7	0.993
EfficientNet-B3 [46]	PCG (sound)	Custom, 1 site	Held-out	Valvular disease	99.35	–
InceptionNet CNN [10]	Heart sound	Public repo.	Held-out	Valvular disease	99.65	–
WbGAS (GAN) [37]	ECG	Custom, GAN-aug.	Held-out	Disease class.	98.2	–
Autoenc+ DenseNet [38]	Clinical	Cleveland (*n*=303)	Held-out	Binary presence	99.67	–
XAI-HD [24]	Clinical	CHD/FHD/SHD (small sets)	Cross-val.	Binary presence	97.5–99.0*	0.98
SGO-XGB/RF [14]	Clinical	Cleveland (*n*=303) / Statlog (*n*=270)	Cross-val.	Binary presence	93.4 / 97.6	0.95 / 0.98
MLDCNN+ AEHOM [26]	Clinical+ ECG	Cleveland+3 others, pooled	80/10/10	Binary presence	96.0 train / 88.0 test	–
CNN Multimodal [47]	BMI+ ECG+ PTB	Multiple sets, pooled	Held-out	Binary presence	98.0	–
Voting Ensemble [39]	Clinical	2 sets, pooled	Held-out	Binary presence	98.41	–
EchoNet CNN [21]	Echo video	EchoNet-Dyn. (*n*=10,030)	External	LVEF (regression)	MAE 4.1%	–

*XAI-HD reports a 20–25% relative reduction in classification error over baseline ML models across the CHD/FHD/SHD sets rather than a single point accuracy; the range shown is derived from that reported error reduction and should be treated as approximate

10.1 Observations

Several observations emerge from the consolidated comparison, and each is stated here together with the caveat needed to interpret it correctly:

1. **ECG-based deep learning reports the highest accuracy figures** (97-–99.7%), but these figures come almost exclusively from single-database, held-out-split evaluations (e.g. MIT-BIH, PTB) rather than from external or multi-centre cohorts, so they indicate performance on that specific benchmark rather than a generalisable clinical accuracy.
2. **Clinical tabular data with ensemble boosting (XGBoost, RF)** typically reports 82–94% accuracy on cross-validated benchmark splits (Table 1), yet several studies in Table 2 report accuracy above 96–99% on the same small ($n < 310$) Cleveland/Statlog datasets. The MLDCNN+AEHOM study [26] is instructive here: it reports 96% training accuracy against only 88% test accuracy on pooled UCI data, an 8-point gap that is a direct, reported illustration of the overfitting and optimistic-bias risk that Reviewer 2 and Reviewer 3 raise; we treat accuracy figures above roughly 95% on datasets of a few hundred records as a signal to check for data leakage, non-independent train/test splits, or small curated test folds rather than as evidence of genuine predictive robustness.
3. **Transformer-based models demonstrate superior data efficiency**— HeartBEiT in particular is explicitly reported to improve over CNN baselines at low sample

sizes—but this advantage is documented mainly for models pre-trained on very large ECG corpora (millions of recordings); it does not establish that transformers are more computationally efficient, since their parameter counts and inference cost are typically higher than CNNs or LSTMs of comparable accuracy (Sect. 11).

4. **Multi-modal fusion outperforms single-modality baselines** in every multi-modal study we reviewed, but the number of such studies is small and none report external validation, so this should be read as a consistent *direction* of effect rather than a precisely quantified gain.
5. **XAI integration does not measurably degrade predictive performance** in the reviewed studies; however, none of them report a prospective, clinician-in-the-loop evaluation of whether the explanations actually change diagnostic decisions or clinician trust (Sect. 11).

11 Discussion: critical synthesis

This section draws together the cross-cutting judgements that a tabular performance comparison alone cannot convey, addressing directly the questions of clinical generalisability, evaluation validity, architecture trade-offs, and translational readiness that are central to interpreting the results in Sect. 10.

11.1 Clinical generalisability of benchmark-trained models

Almost all of the accuracy figures surveyed here, including those exceeding 99%, are obtained on the UCI Cleveland dataset (n=303), the Statlog dataset (n=270), or single-centre ECG corpora such as MIT-BIH and PTB. These datasets were collected at a small number of institutions, from specific patient populations, using specific acquisition equipment and protocols from the 1980s–1990s (for Cleveland/Statlog) or from limited geographic settings (for the ECG corpora). None of the studies in Table 2 report evaluation on an independent cohort drawn from a different institution, country, or patient demographic than the one used for training. Consequently, the benchmark accuracies in this survey should be interpreted as an upper bound on in-distribution performance, not as an estimate of how these models would perform on the demographically and institutionally diverse populations encountered in real-world healthcare delivery. We regard the near-complete absence of external, multi-site, or prospective validation as the single most important limitation of the reviewed literature, and we have moderated the language used elsewhere in this survey (abstract, conclusion, and Table 2) accordingly.

11.2 Distinguishing robustness from overfitting, leakage, and bias

Several of the >99% accuracy figures reported in the surveyed literature are difficult to reconcile with the intrinsic difficulty of heart disease classification from a few hundred tabular records, and typically trace to one or more of the following: (i) non-independent train/test splits, where oversampling (e.g. SMOTE) is applied before rather than after the split, leaking synthetic near-duplicates of test cases into training; (ii) repeated hyperparameter tuning against the same held-out test fold, which inflates the reported figure for that specific fold; and (iii) small, manually curated test sets rather than a fully independent cohort. The MLDCNN+AEHOM study [26] is a useful illustrative case because it explicitly reports both training (96%) and test (88%) accuracy on the same pooled dataset; the 8-point gap is consistent with a degree of overfitting and is more informative

than the training figure alone. As a general heuristic for readers of this survey, we recommend treating any accuracy above roughly 95% on a dataset of a few hundred records with scepticism until the paper's cross-validation and leakage-control procedure has been checked, and we have applied that scepticism throughout Sect. 10.

11.3 Feature engineering and interpretability trade-offs

Among the feature selection techniques discussed in Sect. 4, filter-based methods (correlation and mutual-information ranking, chi-squared testing) contribute the least to predictive performance but the most to interpretability, since the retained features map directly onto established clinical risk factors (age, chest pain type, ST depression, number of major vessels). Wrapper methods (recursive feature elimination) and embedded methods (tree-based importance, L1-regularised regression) recover a small additional accuracy margin (typically 1–3 percentage points) at some cost to interpretability, because the selected feature subset can vary between resampling folds. Metaheuristic selection (genetic algorithms, particle swarm optimisation) tends to produce the largest accuracy gains but the least stable and least clinically interpretable feature subsets. Overall, embedded tree-based importance combined with a post-hoc explainer (SHAP) offers the best-documented balance between predictive performance and interpretability in the studies we reviewed.

11.4 Architecture trade-offs: CNNs, LSTMs, and transformers

CNNs remain the most computationally efficient architecture for fixed-length ECG segments and are the most widely deployed on resource-constrained hardware. LSTMs and other recurrent architectures add modelling capacity for beat-to-beat temporal dependencies at a higher training cost and are more prone to vanishing-gradient issues on long sequences, which hybrid CNN-LSTM/BiLSTM designs partially mitigate. Transformer-based models report the strongest data efficiency at low sample sizes when pretrained on very large corpora (as with HeartBEiT [32]), but this advantage is contingent on access to large-scale pre-training data that most institutions lack, and transformers generally carry higher parameter counts and inference latency than comparable CNNs, which works against the real-time and edge-deployment requirements discussed in Sect. 12. No single architecture dominates on all three axes of accuracy, computational efficiency, and data efficiency simultaneously; the appropriate choice is dataset-size- and deployment-context-dependent rather than universal.

11.5 Does XAI improve clinical trust? A cautious answer

SHAP, LIME, and Grad-CAM are reported, across the studies we reviewed, to improve the *plausibility* of model output to a technically literate reader by highlighting feature contributions or salient image regions that align with known clinical risk factors. However, none of the reviewed studies report a prospective or clinician-in-the-loop evaluation measuring whether these explanations change diagnostic decisions, calibrate clinician trust appropriately, or improve patient outcomes; the evidence for a clinical-trust benefit is therefore indirect (face-validity of the explanations) rather than outcome-based. Post-hoc explanations also do not verify that a model's internal reasoning is causally aligned with clinical pathophysiology, and a plausible-looking explanation can

coexist with a spuriously learned shortcut. XAI should accordingly be regarded as a necessary but not sufficient condition for clinical trust.

11.6 Multimodal fusion: benefits and integration challenges

Integrating heterogeneous modalities (ECG, imaging, EHR, laboratory values, wearable streams) into a single predictive framework raises several technical difficulties beyond model architecture: differing sampling rates and temporal resolutions across modalities; missing modalities for a subset of patients, which forces either imputation or modality-dropout training; heterogeneous feature scales and semantics across institutions and EHR systems; and the risk that a fusion model implicitly relies on whichever modality is least noisy in the training data, which may not hold at deployment. Despite this, every multimodal study reviewed here reports higher accuracy than its single-modality counterpart, which is consistent with fusion providing genuine complementary information; the magnitude of that benefit, however, is not established with the same rigour as the qualitative direction, since sample sizes for multimodal studies remain small and none report external validation.

11.7 Fairness, regulatory readiness, and federated learning

Bias-mitigation approaches identified in the reviewed literature include re-weighting or re-sampling training data by demographic subgroup, adversarial de-biasing (training a secondary discriminator to remove demographic information from learned representations), and post-hoc threshold adjustment per subgroup; audits such as that of Straw et al. [48] indicate that these techniques reduce but do not eliminate differential performance across sex, age, and race, and that subgroup performance is rarely reported at all in the underlying prediction studies. On regulatory readiness, current systems remain distant from clinical-grade deployment under FDA/EMA/CE frameworks: regulatory pathways for AI as a Medical Device require documented external validation, a defined intended-use population, and post-market surveillance for model drift, none of which are demonstrated by the largely single-dataset, retrospective evaluations reviewed here. Federated learning is a promising route to multi-institutional training without centralising patient data, but the reviewed federated studies [45] report that performance degrades under non-independent-and-identically-distributed (non-IID) data across sites (e.g. differing disease prevalence or feature distributions between hospitals), and that differential-privacy noise added to protect patient confidentiality trades off directly against predictive accuracy; neither limitation is yet fully resolved in the literature.

11.8 Class imbalance: what actually works

Across the reviewed studies, SMOTE and its variants (ADASYN, SMOTETomek, SMOTEENN) are the most commonly applied and most reliably reported to improve recall on the minority (disease-positive or rare-arrhythmia) class, provided that resampling is applied strictly within the training fold (see Sect. 11.2). Cost-sensitive learning (class-weighted loss functions) is reported as a lighter-weight alternative that avoids synthetic-sample leakage risk entirely, at a typically smaller improvement in minority-class recall. GAN-based augmentation [37] is reported to generate more diverse synthetic minority samples than SMOTE for signal data, but at higher computational cost and with a risk of generating physiologically implausible waveforms if not carefully validated. No

technique fully eliminates the false-positive-rate cost of improving minority-class recall; the reviewed studies consistently show a precision/recall trade-off rather than a free improvement on both axes.

11.9 Which emerging paradigm has the greatest near-term potential?

Considering translational feasibility rather than technical novelty alone, we judge multimodal fusion and federated learning to have the greatest near-term (3–5 year) potential for advancing cardiovascular diagnostics, because both build on data and infrastructure that health systems already generate (routine EHR, ECG, and imaging data) and both directly target the generalisability gap identified in Sect. 11.1. Foundation models pretrained at very large scale (Sect. 13) have the greatest long-term potential but currently depend on access to proprietary, large-scale ECG/imaging corpora that remain concentrated in a small number of institutions; digital twins and causal AI remain earlier stage, with few clinically validated cardiac applications to date. This ranking reflects our synthesis of the reviewed literature and is offered as an assessment rather than a consensus finding, since the field has not converged on this question.

12 Challenges and limitations

12.1 Data scarcity and imbalance

A persistent challenge in cardiac ML research is the scarcity of large, annotated, multicentre datasets. The UCI Cleveland dataset (n=303) is too small for reliable deep learning training. Pathological classes (e.g., rare arrhythmias, congenital defects) are vastly underrepresented relative to normal controls, causing classifiers to exhibit high sensitivity at the expense of specificity or vice versa. SMOTE, GAN-based augmentation, and transfer learning mitigate but do not eliminate this problem.

12.2 Generalisability and external validation

Models trained on a single-centre dataset often fail to generalise to different populations, ethnicities, scanner types, or clinical workflows. The majority of published studies evaluate models only on held-out test sets from the same distribution, without prospective or external validation. Multi-site datasets (MIMIC, PhysioNet) and federated learning are promising solutions.

12.3 Reproducibility

Many published accuracy claims are difficult to reproduce due to: incomplete code and data sharing, undisclosed preprocessing decisions, test-set contamination, and hyperparameter overfitting to small validation sets. The field would benefit from standardised evaluation protocols, pre-registered experiments, and open-source benchmarks.

12.4 Missing and heterogeneous data

Clinical datasets routinely contain missing values, inconsistent measurement units, and heterogeneous feature definitions across institutions. Data integration from EHRs, wearables, and imaging systems requires sophisticated harmonisation pipelines.

12.5 Model interpretability and regulatory compliance

Regulatory bodies (FDA, EMA, CE marking) impose requirements on the validation, documentation, and monitoring of AI as a medical device (AIaMD). Current XAI methods provide post-hoc explanations but do not guarantee that model reasoning aligns with clinical causality. Prospective studies evaluating whether XAI improves clinical outcomes are still largely absent.

12.6 Fairness and bias

Several studies have identified demographic bias in cardiac prediction models, with models performing differentially across sex, age, race, and socioeconomic strata. Straw et al. identified fairness issues in publicly available cardiovascular risk models. Developing fairness-aware training objectives and bias auditing pipelines is an active research area [48].

12.7 Real-time and edge deployment

Deploying high-accuracy DL models (e.g., transformers) on resource-constrained wearable or edge devices requires model compression without significant accuracy loss. Balancing model complexity with real-time inference latency remains an engineering challenge.

12.8 Multi-label and comorbid disease classification

Patients often present with simultaneous multiple cardiac pathologies. Most existing classifiers assume mutually exclusive labels (single diagnosis), whereas clinical reality requires multi-label classification. Few models currently address the concurrent detection of multiple abnormalities on the same ECG recording [49].

13 Future research directions

13.1 Foundation models and large-scale pre-training

Following the success of HeartBEiT [32] and analogous NLP foundation models (GPT, BERT), the cardiological AI field is moving towards large-scale pre-training of transformer models on tens of millions of ECGs, echocardiograms, and imaging data. Such models would exhibit strong zero-shot or few-shot generalisation to rare cardiac conditions.

13.2 Multi-modal fusion and holistic cardiac profiles

Future systems will integrate ECG signals, cardiac MRI, echocardiography, clinical laboratory values, genomic data, and real-time wearable streams into unified, multi-modal deep learning models. Cross-modal attention transformers and graph neural networks (GNNs) encoding patient similarity networks represent promising architectures.

13.3 Causal inference and counterfactual explanations

Shifting from correlational ML models to causal frameworks will enable clinicians to reason about interventions ("if this patient's blood pressure were reduced by 20 mmHg, how would cardiovascular risk change?"). Counterfactual explanations generated by causal models offer actionable clinical guidance beyond current XAI methods.

13.4 Continual and online learning

Cardiac models deployed in clinical practice must adapt to patient population drift, updated clinical guidelines, and new data without full retraining. Continual learning methods that resist catastrophic forgetting are critical for maintaining model accuracy over time.

13.5 Digital twins for cardiac simulation

Patient-specific digital twins—computational models of individual cardiac physiology constructed from imaging and clinical data—can simulate the response to therapies and interventions. Integrating ML with biophysical cardiac simulation represents a frontier in precision cardiology.

13.6 Federated and privacy-preserving learning at scale

Scaling federated learning across global health systems, incorporating differential privacy guarantees and secure aggregation protocols, will enable truly population-level cardiac AI while protecting patient confidentiality.

13.7 Prospective clinical trials

The field urgently requires randomised controlled trials evaluating whether AI-assisted cardiac diagnosis improves patient outcomes (reduced mortality, hospitalisation, cost) compared to standard care. Observational studies and retrospective evaluations, while valuable, are insufficient evidence for clinical deployment.

14 Conclusion

This comprehensive survey has examined the landscape of computational methods for heart disease detection, spanning more than a decade of research across diverse data modalities, algorithmic paradigms, and clinical applications. The following conclusions are drawn:

1. **Machine learning shows promise as a clinical decision-support tool** for heart disease risk stratification from structured clinical data. XGBoost, Random Forest, and ensemble boosting methods typically report 82–94% accuracy on standard, single-centre benchmarks, offering a pragmatic balance between predictive power and interpretability; as discussed in Sect. 11.1, this evidence base does not yet establish performance on independent, multi-institutional cohorts.
2. **Deep learning has substantially advanced ECG and cardiac imaging analysis on benchmark datasets**. CNN, LSTM, and transformer architectures report high in-distribution accuracy (up to 97-–99.7%) for arrhythmia classification, valvular disease detection, and LVEF estimation on the specific datasets studied; whether this translates into near-clinical accuracy in routine practice remains to be demonstrated through external and prospective validation (Sect. 11.1).
3. **Explainability is no longer optional**. SHAP, LIME, Grad-CAM, and hybrid XAI frameworks are essential components of clinically deployable systems, enabling transparency, trust, and regulatory compliance.
4. **IoT, wearables, and federated learning are transforming cardiac care from episodic to continuous**, enabling remote early detection in settings previously without access to specialised cardiology.

5. **Significant challenges remain** in data quality, generalisability, reproducibility, fairness, and regulatory validation. Addressing these gaps requires coordinated efforts across data science, clinical medicine, health informatics, ethics, and regulatory science.
6. **The future of cardiac AI** lies in foundation models pre-trained at massive scale, causal reasoning frameworks, multi-modal integration, and prospective clinical trials that translate algorithmic performance into demonstrated patient benefit. As cardiovascular disease burden continues to grow globally, the responsible development and deployment of AI-driven cardiac detection systems holds transformative potential—not only for high-resource healthcare systems but critically for the low- and middle-income countries where 75% of cardiovascular deaths occur and specialist access remains scarce [1].

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Human or animal rights participants
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